Listening Homework 1

Book 3 Unit 6

Questions 1-5 Sentence completion

Write **ONE WORD, ONE NUMBER, OR ONE OF EACH** from the indicated SHORT audio. Do not add % where it is printed.

1. [6.1-S01] **The exhibition is called Discovering Medieval _____ .**

2. [6.1-S01] **The Middle Ages have often been called the _____ Ages.**

3. [6.1-S01] **The guide says the period was full of scientific _____ .**

4. [6.1-S01] **Ideas and technology from several regions were brought to _____ .**

5. [6.1-S01] **Medieval inventions made a great _____ to society.**

Listening Homework 1

Book 3 Unit 6

Questions 6-10 Sentence completion

Write **ONE WORD, ONE NUMBER, OR ONE OF EACH** from the indicated SHORT audio. Do not add % where it is printed.

6. [6.1-S01] **The guide says some medieval inventions are very _____ .**

7. [6.1-S02] **The first fountain pen was made in 953 in _____ .**

8. [6.1-S02] **Before the fountain pen, writers used bird _____ and ink.**

9. [6.1-S02] **The old method often left fingers and clothes _____ with black ink.**

10. [6.1-S02] **The first fountain pen held ink in a small _____ .**

Listening Homework 1

Book 3 Unit 6

Questions 11-15 Single choice

Tick **ONE** answer, A-C, for each question. Use the indicated SHORT audio.

11. [6.1-S02] **What display does the guide locate in Room 11B?**

- ☐ A. a fountain-pen model
- ☐ B. an original pair of spectacles
- ☐ C. a weather balloon

12. [6.1-S03] **When were spectacles invented, according to the guide?**

- ☐ A. In the twelfth century
- ☐ B. During next week's holiday
- ☐ C. In the thirteenth century

13. [6.1-S04] **Which rooms contain works by al-Jazari?**

- ☐ A. Rooms 11A and 11B
- ☐ B. Rooms 12A and 12B
- ☐ C. Rooms in a railway station

14. [6.1-S04] **Where did al-Jazari live?**

- ☐ A. In twelfth-century Turkey
- ☐ B. In ninth-century China
- ☐ C. On a modern space station

15. [6.1-S04] **What is the title of al-Jazari's work mentioned by the guide?**

- ☐ A. The Book of Medieval Poems
- ☐ B. The History of Ocean Currents
- ☐ C. The Book of Knowledge of Ingenious Mechanical Devices

Questions 16-20 Single choice

Tick **ONE** answer, A-C, for each question. Use the indicated SHORT audio.

16. [6.1-S04] **What can visitors see of al-Jazari's devices in the exhibition?**

- ☐ A. Models of some of his devices
- ☐ B. Only his original handwritten letters
- ☐ C. A live tennis match

17. [6.1-S05] **How does the guide describe the form of the crankshaft?**

- ☐ A. As a short wheel
- ☐ B. As a page in a menu
- ☐ C. As a long arm

18. [6.1-S06] **How does the guide initially rank the next invention?**

- ☐ A. As one of the most important medieval inventions
- ☐ B. As a minor object with no historical role
- ☐ C. As a recipe for vegetable soup

19. [6.1-S06] **How broadly does the guide describe the invention's historical influence?**

- ☐ A. It affected only one museum room
- ☐ B. It changed world history in ways difficult to imagine
- ☐ C. It changed the colour of the guide's shoes

20. [6.1-S06] **When was gunpowder invented, according to the guide?**

- ☐ A. In the nineteenth century
- ☐ B. In the ninth century
- ☐ C. During next week's lunch break

Questions 21-23 Multiple choice

Tick **TWO** answers, A-E, for each numbered question. Use the indicated SHORT audio.

21. [6.1-S03] **Which TWO statements describe early spectacles?**

- ☐ A. They were held before the eyes or balanced on the nose.
- ☐ B. They helped people with poor sight read.
- ☐ C. They were fixed permanently behind the ears.
- ☐ D. They were designed to improve hearing.
- ☐ E. They forecast tomorrow's weather.

22. [6.1-S04] **Which TWO features does the guide mention in al-Jazari's book?**

- ☐ A. It lists only ten machines.
- ☐ B. It provides stories without construction guidance.
- ☐ C. It lists one hundred machines.
- ☐ D. It predicts football scores.
- ☐ E. It provides instructions and diagrams for construction.

23. [6.1-S05] **Which TWO details does the guide give about the mechanical clocks?**

- ☐ A. Visitors can see al-Jazari's mechanical clocks.
- ☐ B. The clocks are hidden from visitors.
- ☐ C. The clocks operate with petrol and steam.
- ☐ D. The clocks operate with water and weights.
- ☐ E. The clocks translate legal documents.

Questions 24-25 Multiple choice

Tick **TWO** answers, A-E, for each numbered question. Use the indicated SHORT audio.

24. [6.1-S05] **Which TWO uses of the crankshaft are stated?**

- ☐ A. cleaning medieval spectacles
- ☐ B. watering gardens and fields
- ☐ C. working inside modern car engines
- ☐ D. printing the guide's book
- ☐ E. choosing music for a wedding

25. [6.1-S06] **Which TWO statements introduce gunpowder?**

- ☐ A. It belongs to the four great inventions of Roman culture.
- ☐ B. Its discoverers were trying to improve spectacles.
- ☐ C. It was ordered from a restaurant.
- ☐ D. It belongs to the four great inventions of Chinese culture.
- ☐ E. Its discoverers were seeking a life-extending powder.

Passage A - How Lotus Leaves Inspired Self-Cleaning Surfaces

A A lotus leaf can remain strikingly clean although it grows above muddy water. The clean **appearance** is not produced by polishing. In a study published in 1997, botanists Wilhelm Barthlott and Christoph Neinhuis investigated the surface of sacred lotus and other plants. They asked how leaf structure, water repellency and the weak attachment of dirt worked together. Their research turned an old observation about a plant into a testable explanation of self-cleaning.

B A microscope **image** shows that the leaf is not truly flat. Its surface has a fine relief formed partly by wax crystals. The researchers compared microscopically smooth leaves with rough, water-repellent leaves under controlled **conditions**. They placed different particles on the plants and then used artificial rain from a sprinkler or fog generator. This comparison separated the effect of roughness from the simple fact that water had touched a leaf.

C On a rough, water-repellent leaf, a drop touches less of the surface than it would on a smooth leaf. Dirt also attaches less strongly. As the drop rolls away, its **movement** collects loose particles. The study found that rolling drops removed the artificial contamination from the water-repellent leaves. Both parts of the mechanism are **essential**: water must form moving drops, and particles must attach more strongly to the water than to the leaf.

Passage A - How Lotus Leaves Inspired Self-Cleaning Surfaces

D Biomimicry does not require engineers to manufacture a plastic lotus leaf. They can **translate** the principle by combining a water-repellent material with a carefully structured surface. The University of Bonn reports that this research influenced a **range** of work in paints, varnishes and other surface technologies. The useful idea is the relationship among structure, water and particles, rather than the colour or exact shape of the plant.

E The experiment can also **illustrate** why a natural model needs measurement. A smooth leaf may shed water without cleaning in the same way, while a rough surface may fail if its chemistry lets water spread into a film. Researchers also have to compare a new surface with an untreated one, because a dramatic photograph alone cannot measure cleaning. Designers therefore have to **rethink** the principle for each material and use. They test the direction of water, the type of dirt and the durability of the surface instead of assuming that one pattern will work everywhere.

F The result is powerful but not **unlimited**. The 1997 tests concerned particles placed on leaves and removed by controlled rinsing. They did not prove that every stain disappears, that every manufactured coating lasts indefinitely, or that a self-cleaning surface never needs care. Long-term use introduces wear that a short leaf experiment did not examine. This boundary makes the example more useful, not less. The lotus supplied a mechanism to investigate, and engineering decides whether that mechanism survives on a wall, a fabric or another working surface.

Reading Homework 1

Book 3 Unit 6

Questions 1-10 Summary completion

Write **ONE WORD** from the indicated paragraphs. Questions 1-5: **A, B**. Questions 6-10: **C, D**.

A lotus leaf stays clean even while growing above **(1)** _____ water. The article directly rules out **(2)** _____ as the cause of the clean appearance. The researchers examined the weak **(3)** _____ of dirt as part of the mechanism. The plant observation became a testable **(4)** _____ of self-cleaning. Part of the fine relief came from wax **(5)** _____ .

Dirt attaches less **(6)** _____ to the leaf. Loose **(7)** _____ attach more firmly to water than to the leaf. Engineers do not have to **(8)** _____ a plastic lotus leaf. The useful idea is the **(9)** _____ among structure, water and loose dirt. The research influenced paints, **(10)** _____ and other surface technologies.

Questions 11-15 List of Headings

Match each specified paragraph with its best heading. Tick **ONE** box. Some headings are not used.

i. The economic history of flower markets

ii. Turning a plant observation into a research question

iii. Transferring a natural mechanism into technology

iv. How drops remove loosely held dirt

v. Comparing leaf surfaces under controlled rain

vi. A guarantee that every stain will disappear

vii. Testing whether an adaptation really works

11. Paragraph A

☐ i ☐ ii ☐ iii ☐ iv ☐ v ☐ vi ☐ vii

12. Paragraph B

☐ i ☐ ii ☐ iii ☐ iv ☐ v ☐ vi ☐ vii

13. Paragraph C

☐ i ☐ ii ☐ iii ☐ iv ☐ v ☐ vi ☐ vii

14. Paragraph D

☐ i ☐ ii ☐ iii ☐ iv ☐ v ☐ vi ☐ vii

15. Paragraph E

☐ i ☐ ii ☐ iii ☐ iv ☐ v ☐ vi ☐ vii

Questions 16-20 Single choice

Tick **ONE** answer, A-C, for each question.

16. When was the reported plant-surface study published?

- ☐ A. During the first moon landing
- ☐ B. In 2022
- ☐ C. In 1997

17. How did the team apply artificial rain?

- ☐ A. With a sprinkler or fog generator
- ☐ B. By tuning a radio transmitter
- ☐ C. By placing leaves in a river

18. What did comparing the leaves help separate?

- ☐ A. The effect of roughness from simple water contact
- ☐ B. The colour of lotus flowers from their age
- ☐ C. The speed of a passenger train

19. Why can a dramatic photograph be insufficient evidence?

- ☐ A. It always changes the chemistry of the surface
- ☐ B. It cannot measure how well a surface cleans
- ☐ C. It cannot show the score of a tennis match

20. Why might a smooth leaf shed water without cleaning in the same way?

- ☐ A. Smoothness guarantees removal of every stain
- ☐ B. Water loss alone does not provide the full tested mechanism
- ☐ C. The leaf must first be painted blue

Questions 21-25 Single choice

Tick **ONE** answer, A-C, for each question.

21. What can make a rough surface fail?

- ☐ A. Its owner changes a bicycle tyre
- ☐ B. It is examined with an untreated comparison
- ☐ C. Its chemistry lets water spread into a film

22. Why compare a new surface with an untreated one?

- ☐ A. To judge whether the treatment improves cleaning
- ☐ B. To make both surfaces look identical
- ☐ C. To calculate a restaurant bill

23. Which set of variables does Paragraph E say designers test?

- ☐ A. Water direction, dirt type and durability
- ☐ B. Flower colour, seed size and sunlight
- ☐ C. Ticket price, platform number and timetable

24. Which issue was outside the short leaf experiment?

- ☐ A. How to schedule a school orchestra
- ☐ B. Wear during long-term use
- ☐ C. Whether controlled rinsing removed added dirt

25. Which conclusion best represents the article?

- ☐ A. One leaf experiment guarantees permanent cleaning on every product
- ☐ B. Lotus leaves explain how to repair computer passwords
- ☐ C. Nature supplies a mechanism that engineering must test for each use

Speaking Homework 1

Book 3 Unit 6

Questions 1-5 Build sentences

Put the word groups in order. Use each group once. Do not add words or change word forms. Record each full sentence.

1. Start with "The model".
The model, of, is made, cardboard

2. Start with "The first design".
by, our class, was tested, The first design

3. Start with "This tool".
makes, the task, easier, This tool

4. Start with "The parts".
with, The parts, small screws, are joined

5. Start with "The app".
new students, for, The app, was created

Questions 6-10 Complete the replies

Read each situation and follow its clue to complete the reply. Record the whole reply.

6. Situation: The model is cardboard. Choose of or by.

Friend: What is it made of?

You: It is made _____ cardboard.

7. Situation: The class tested it yesterday. Choose was or is.

Friend: Has anyone tested the model?

You: Yes, it _____ tested yesterday.

8. Situation: The new handle helps you carry the bag. Choose makes or does.

Friend: Why did you add that handle?

You: It _____ the bag easier to carry.

9. Situation: You changed one part of a design. Choose did or made.

Friend: Is this exactly the same design?

You: No, I _____ one change.

10. Situation: Your group created the app. Choose with or by.

Friend: Who created it?

You: It was created _____ our group.

Speaking Homework 1

Book 3 Unit 6

Questions 11-15 Order and speak

Put sentences A-C in order. Record the opening, then read all three sentences in the correct order. Do not read the letters.

11. Describe a model's construction.

Opening: Our model started with very simple materials.

A	The finished model was then placed on a stand.
B	Those circles were joined with a short tube.
C	Two pieces of cardboard were cut into circles.

12. Explain a useful change.

Opening: We wanted the box to be easier to carry.

A	That handle gave the user a better grip.
B	That better grip made lifting the box easier.
C	A handle was added to the top.

13. Describe testing a design.

Opening: We checked the first version before showing it.

A	Their comments showed that the base was too narrow.
B	In response to those comments, a wider base was made for the next version.
C	The model was tested by three classmates.

14. Describe an app project.

Opening: Our group made something for new students.

A	Using those classroom locations, new students found their first lesson.
B	We created an app with a school map.
C	That map showed the main classrooms.

15. Explain a material choice.

Opening: We needed a material we could shape easily.

A	We chose the shape that stood most steadily.
B	That paper was folded into several different shapes.
C	The first model was made from paper.

Speaking Homework 1

Book 3 Unit 6

Questions 16-18 Follow the model

Use the model to make your own answer in 3-4 connected sentences. Write key words, then record your answer.

16. Describe an invention made from familiar materials. Use a passive form.

Model: Our phone stand is made from cardboard. Its base is folded into a triangle. The stand holds a phone upright.

17. Describe one change to an everyday object. Use make.

Model: We added a handle to the box. The handle makes it easier to carry. It is made from strong fabric.

18. Describe a classroom test. Say who tested what and one result.

Model: The paper bridge was tested by our group. We placed two small books on it. The bridge stayed upright.

Speaking Homework 1

Book 3 Unit 6

Questions 19-20 Follow the model

Use the model to make your own answer in 3-4 connected sentences. Write key words, then record your answer.

19. Describe a simple app you could create and who it would help.

Model: Our app could show the school timetable. It would be designed for new students. It could make planning the day easier.

20. Explain how a familiar tool is made useful by one part.

Model: A zip has a small handle. The handle is used to pull the two sides together. It makes the bag easier to close.

Speaking Homework 1

Book 3 Unit 6

Questions 21-22 Describe the picture

Write key words, then record 2-3 sentences for each question. Use the words if helpful.



Words: device, user, allow

21. Describe the phone stand. Say what it is made from and how it helps the user.

Words: make attempts, product, outcome

22. Describe what the two students are doing. Suggest one test they could make.

Speaking Homework 1

Book 3 Unit 6

Questions 23-25 Describe the picture

Write key words, then record 2-3 sentences for each question. Use the words if helpful.



Words: install, device, user

23. Describe the library device and the person using it. Use one passive form.

Words: access, allow, make a difference

24. Imagine you work in this library. Explain one way you could help this visitor.

Words: make improvements, device, user

25. Suggest one improvement to the library screen. Explain its possible benefit.

Writing Homework 1

Book 3 Unit 6

Questions 1-5 Build sentences

Put the word groups in order. Use each group once. Do not add words or change word forms. Write each full sentence.

1. Start with "The new app".

may, The new app, our class, help

2. Start with "This design".

change, will probably, This design, next year

3. Start with "The first model".

is, The first model, to work perfectly, unlikely

4. Start with "We need".

the damaged, to rebuild, We need, frame

5. Start with "The next version".

could, a larger screen, include, The next version

Questions 6-10 Complete the sentences

Complete each sentence. Follow the word choices or clues in each question.

6. Choose is or may.

The new tool _____ help us work faster.

7. Choose probably or probable.

The next model will _____ be smaller.

8. Add re- to build to mean build again.

The students will _____ the model.

9. Choose unlikely or likely. The team expects a low chance of success.

The first attempt is _____ to succeed.

10. Choose unable or enable.

The new feature could _____ users to share notes.

Questions 11-15 Choose the sentence order

Read sentences A-C. Tick one sequence for each question.

11. Order a cautious prediction.

A Its smaller size may make it easier to carry.
B The group is designing a smaller lamp.
C More testing is needed to check that possible benefit.

☐ B-A-C ☐ A-B-C ☐ B-C-A

12. Order an account of rebuilding.

A The model broke during the first test.
B The rebuilt version has a stronger base.
C The team decided to rebuild it.

☐ A-C-B ☐ B-C-A ☐ C-A-B

13. Order a prediction about an app.

A That feature might help students plan their study time.
B The class may create a homework app.
C The app could display the next task for each subject.

☐ B-A-C ☐ B-C-A ☐ A-B-C

14. Order a design revision.

A The first instructions were unclear.
B The new wording may help future users understand them.
C To address this problem, the group rewrote the difficult steps.

☐ B-C-A ☐ C-A-B ☐ A-C-B

15. Order a reason for caution.

A Only one version of the device has been tested.
B Its performance outside that classroom is still uncertain.
C That version worked in a quiet classroom.

☐ A-B-C ☐ A-C-B ☐ C-A-B

Questions 16-18 Follow the model

Use the model to make your own answer in 3-4 connected sentences. Write your own answer with different details.

16. Predict a possible use of a classroom app. Give a reason.

Model: A class app could display homework dates. Students may find it easier to plan their week. They would have the information in one place.

17. Describe a failed model and predict a change. Use rebuild.

Model: The paper tower fell during testing. We will rebuild it with a wider base. The new version may stand more steadily.

18. Make a cautious prediction about a smaller version of an invention.

Model: A smaller lamp will probably be easier to carry. It may fit inside a school bag. However, its light could cover a smaller area.

Questions 19-20 Follow the model

Use the model to make your own answer in 3-4 connected sentences. Write your own answer with different details.

19. Describe an unclear instruction and a possible improvement. Use a word with re-.

Model: The first guide uses long sentences. We could rewrite them in simpler language. The new guide may be easier to follow.

20. Explain why an invention might not work for everyone. Use unlikely.

Model: An app is unlikely to help someone without a suitable device. That person may need a printed guide. The designer should consider both options.

Writing Homework 1

Book 3 Unit 6

Questions 21-22 Describe the picture

Write 2-3 sentences for each question. Use the words if helpful.



Words: disorganized, personal, essential

21. Describe the cooking scene. Make one prediction using may or might.

Words: disorganized, rethink, personal

22. Imagine the cook often loses track of time. Predict how a phone timer could change her routine.

Writing Homework 1

Book 3 Unit 6

Questions 23-25 Describe the picture

Write 2-3 sentences for each question. Use the words if helpful.



Words: appearance, personal, range

23. Describe the people and the radio. Say what might happen next.

Words: personal, rethink, essential

24. Imagine the radio has been repaired. Predict one benefit of using it again.

Words: unsafe, conditions, essential

25. Imagine you are the radio owner. Write one question about its condition and one possible next step.